

Coverage, Not One Fake

DATA 443/643 · Week 11, note 11.3

Week 11

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Ideas from Stanford CS231N 2025 L13 (implicit samples; you still need coverage). Original slides; that course is not copied.

Lecture plan

- 1 D only fights the support of G
- 2 Fidelity versus coverage
- 3 92/8 is collapse; 47/53 is covered
- 4 Write the check before you train

Key goal: sharp and identical is collapse, not success.

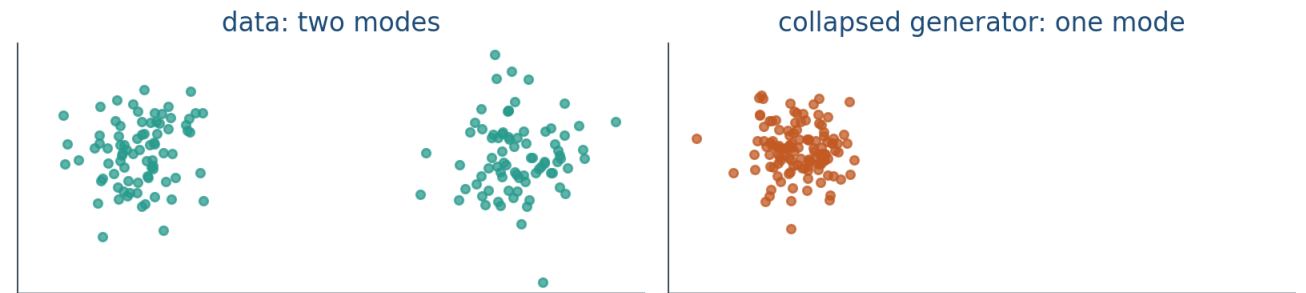
First time: what / why / formula

- What / why. One pretty sample is not a trained G . Coverage: 92/8 on two Gaussians is collapse; 47/53 is covered. D only scores “is this fake?” It does not have a missing-mode term. G can fool D with one spike.
- Architecture. Same GAN. Evaluation needs bins / modes, not only loss_G .
- Formula. There is no extra loss term in vanilla J that says “visit mode 2.” That is the bug.
- Tradeoffs. + Easy to demo. – Image metrics (FID) are imperfect coverage proxies; fixing collapse is still research-level.

How. Lab 11: two 1-D Gaussians. Count mass per mode. Fixes (brief): more diverse z , unrolled/regularized D , or switch to diffusion (Week 12).

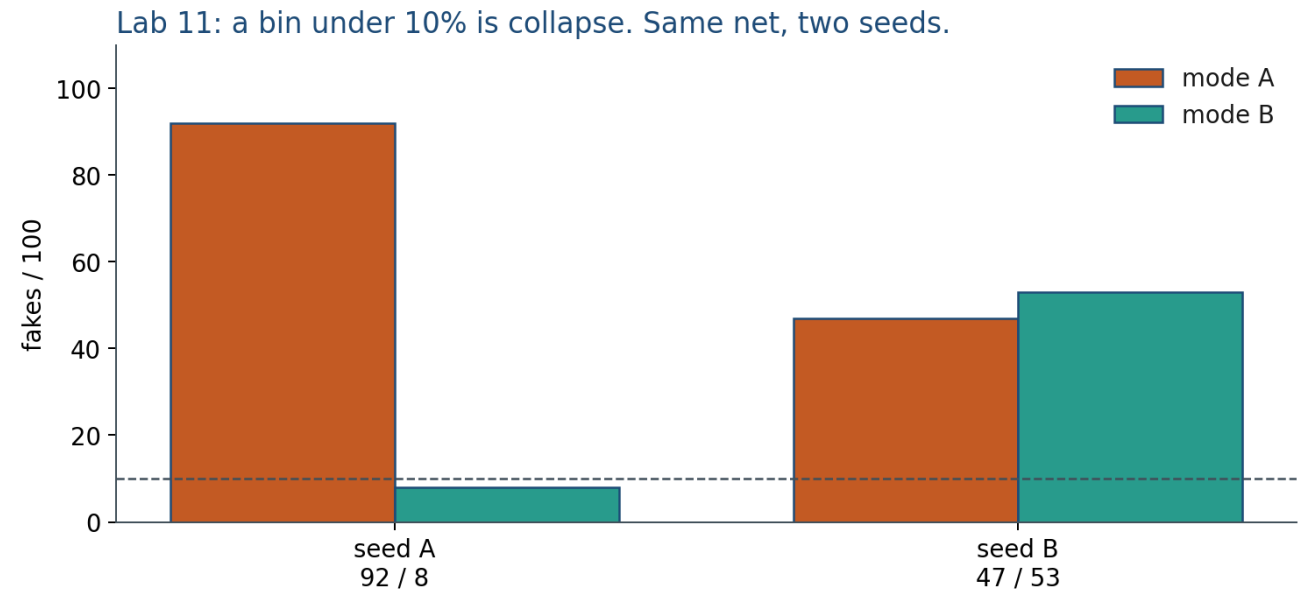
One mode is enough to fool D

- G parks on cluster A. D learns B is real.
- G may jump to B, or never leave A.
- Modes G never visits do not appear in G's loss.



Two seeds, same architecture

- 100 fakes. Nearest-mean bins.
- 92 / 8: the 8% bin is under 10%. Collapse.
- 47 / 53: write covered. Report both fractions.



Two questions

Fidelity

Does this fake look real? A grid of images. Precision.

Coverage

Did we hit the real support? Recall. FID is a proxy with a large sample.

Why text-to-image moved

GAN

One forward pass. Fast. Collapse is the usual failure.

Diffusion (Week 12)

T reverse steps. Slower. Usually covers better. The coverage lesson does not move.

What to take from this lecture

Bin rule

Either mode under 10% of 100 fakes: collapse.

Loss is silent

G's loss has no term for a missing cluster.

Lab 11

Print the two percentages, not only lossG.